

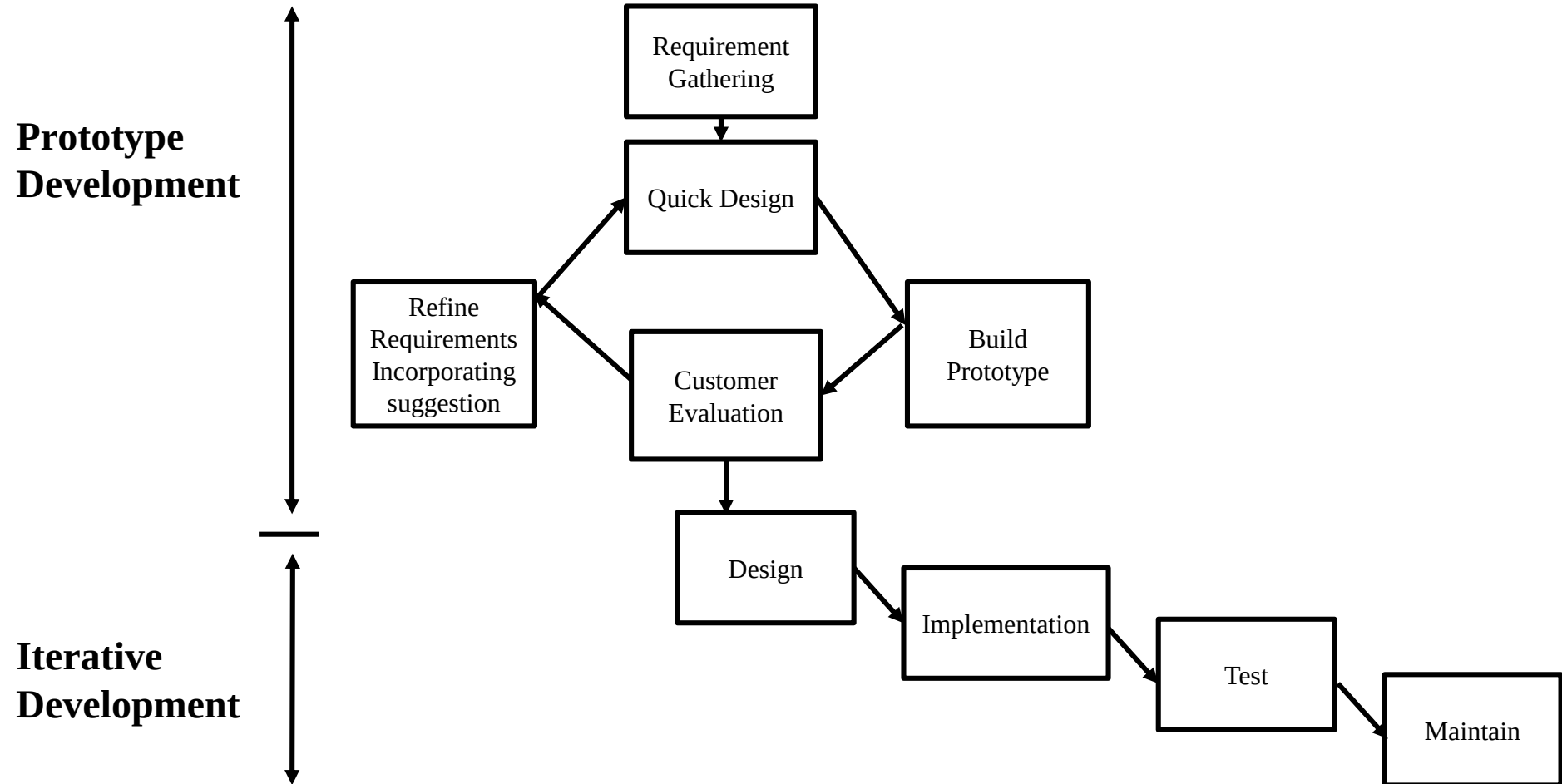
CSE327

Software Engineering

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LECTURE - 4

Prototype Model



Prototype Model

What is prototype model?

- ❑ A prototype model is a toy/demo implementation of the actual product or system.
- ❑ A prototype model usually exhibits limited functional capabilities, low reliability, and inefficient performance as compared to the actual software.
- ❑ A prototype model is usually built using several shortcuts. The shortcuts might involve using inaccurate, inefficient or dummy functions.
- ❑ A prototype usually turns out to be a very crude version of the actual system.

Prototype Model

- ❑ In the prototype model, prototyping starts with initial requirements gathering phase.
- ❑ Quick design is carried out and a prototype is built. The developed prototype is submitted to the customer for his assessment.
- ❑ Based on the customer feedback, the requirements are refined, and the prototype is suitably modified.
- ❑ This cycle of obtaining customer feedback and modifying the prototype continues until the customer approves the prototype.
- ❑ Once the customer approves the prototype, the actual system is developed using the iterative waterfall approach.

Prototype Model

Why do we need Prototype Model?

- ❑ To illustrate the input data formats, messages, reports, and interactive dialogues to the customer.
- ❑ To gain a better understanding of the customer's needs: how the screen might look like, how the user interface would behave, and how the system would produce the output?
- ❑ To examine the technical issues associated with product development.
- ❑ It is not possible to get the perfect product in the first attempt.
- ❑ Many engineers and researchers advocate that if you want to develop a good product you must plan to throw away the first version.

Prototype Model

Advantage:

- ❑ Demo working model: Customer get demo working model of actual product which help them to give a better understanding and attain a high level of satisfaction.
- ❑ New requirement: Based on the customer feedback, the requirements are redefined, and the prototype is suitably modified till final approval.
- ❑ Missing functionality: can be easily established.
- ❑ Easy error detection: It saves time and cost in developing the prototype and enhance the quality of the final product.
- ❑ Flexibility: in the development phase.

Prototype Model

Disadvantage:

- ❑ Time-consuming: As the prototype is being modified time to time according to customer requirement which usually increases the time of completion of the product.
- ❑ Complexity: Change in the requirement usually expand the scope of the product beyond its original plan and thus increase the complexity.
- ❑ Poor Documentation: Continuous changing of requirement can lead to poor documentation.
- ❑ Unpredictability of no of iteration: It is difficult to determine the no of iteration required before the prototype is finally accepted by the customer.
- ❑ Confusion: Customer can confuse between the actual product and prototype.

Requirement Engineering

What is Requirement Engineering?

- ❑ Requirements engineering ([RE](#)) refers to the process of defining, documenting, and maintaining requirements in the engineering design process.
- ❑ Requirement engineering provides the appropriate mechanism to understand:
 - What the customer desires [types of software, interface]
 - Analyzing the need [What we need]
 - Assessing the feasibility [Economic feasibility, technical, operational]
 - Negotiation reasonable solution [Prioritize the modules]

Requirement Engineering

- ❑ Specifying the solution clearly [What we have and not]
- ❑ Validating the specification
- ❑ Managing the requirement

Requirement Engineering

Types of Requirement

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graph TD; A[Types of Requirement] --> B[User Requirement]; A --> C[System Requirement];
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User Requirement

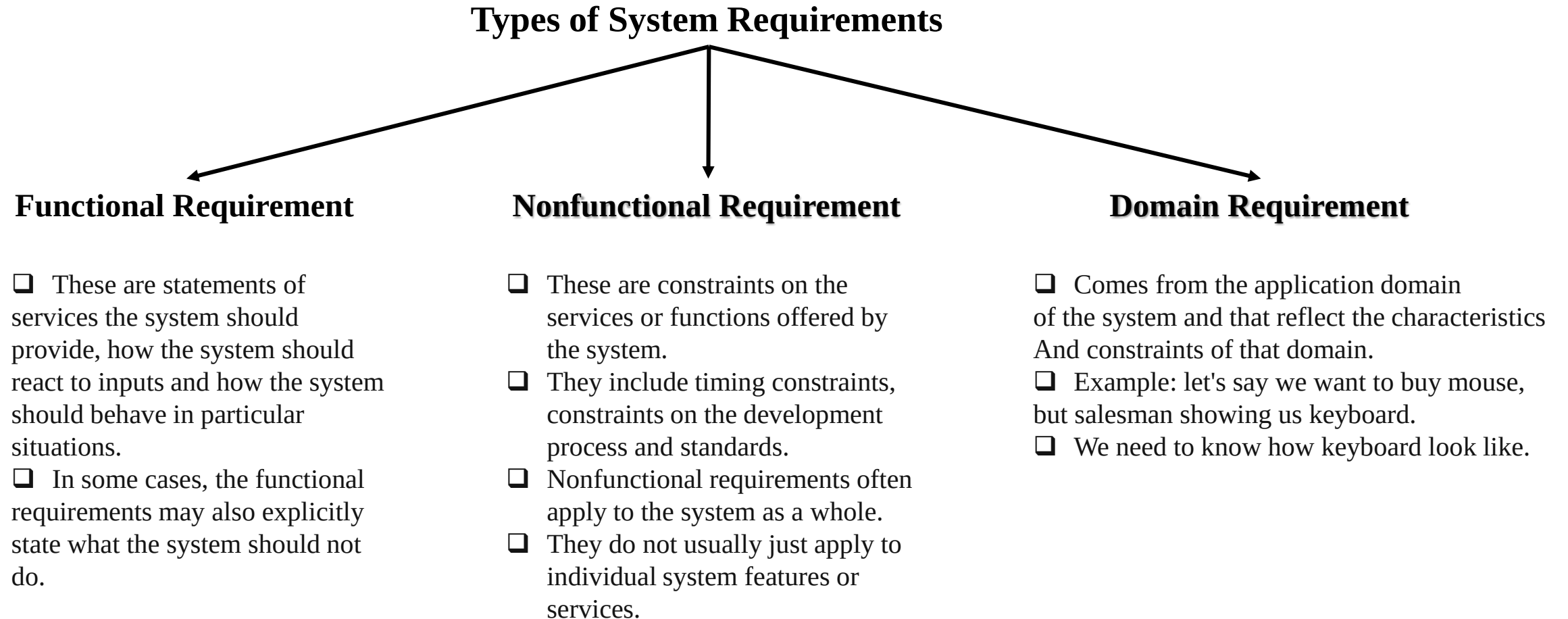
- ❑ User requirements are statements, in a natural language plus diagrams, of what services the system is expected to provide and the constraints under which it must operate.

System Requirement

- ❑ System requirements set out the system's functions, services and operational constraints in detail. The system requirements document (sometimes called a **functional specification**) should be precise.
- ❑ It should define exactly what is to be implemented.
- ❑ It may be part of the contract between the system buyer and the software developers.

System Requirements

Types of System Requirements



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graph TD; A[Types of System Requirements] --> B[Functional Requirement]; A --> C[Nonfunctional Requirement]; A --> D[Domain Requirement];
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Functional Requirement

- ❑ These are statements of services the system should provide, how the system should react to inputs and how the system should behave in particular situations.
- ❑ In some cases, the functional requirements may also explicitly state what the system should not do.

Nonfunctional Requirement

- ❑ These are constraints on the services or functions offered by the system.
- ❑ They include timing constraints, constraints on the development process and standards.
- ❑ Nonfunctional requirements often apply to the system as a whole.
- ❑ They do not usually just apply to individual system features or services.

Domain Requirement

- ❑ Comes from the application domain of the system and that reflect the characteristics And constraints of that domain.
- ❑ Example: let's say we want to buy mouse, but salesman showing us keyboard.
- ❑ We need to know how keyboard look like.